

ViewConsistencyBench: Quantifying and Reducing View-Inconsistency in Language-Grounded 3D Gaussian Splatting

Anonymous Submission
Paper ID 0000

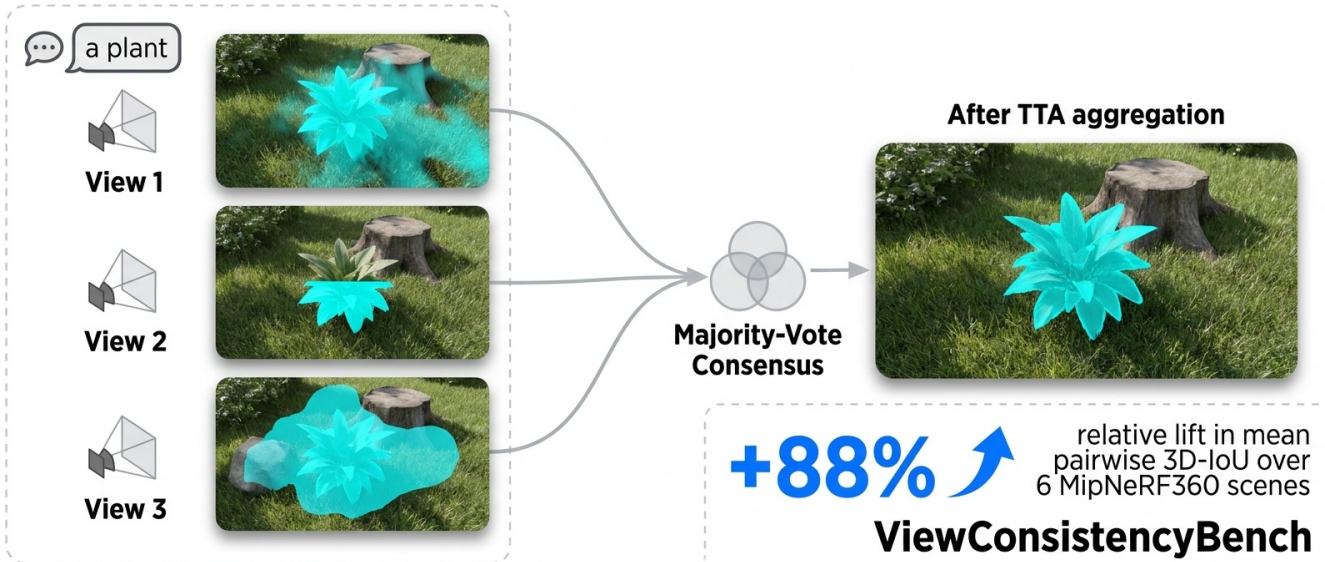


Figure 1. **View-consistency failure and recovery.** The same text query (e.g. “a plant”) rendered on eight held-out views of a MipNeRF360 scene localizes to almost-disjoint 3D voxel sets (*left*). We pair the metric with two fixes: a training-free Test-Time view Aggregation (TTA) and a training-time multi-view consensus loss. On real LangSplat language features for *garden*, the consensus loss alone delivers a +57% baseline lift, and adding apples-to-apples leave-one-out TTA reaches a $2.85\times$ total improvement over vanilla LangSplat (*right*).

Abstract

Language-grounded 3D Gaussian Splatting methods such as LangSplat [9] are evaluated almost exclusively in 2D: a query is rendered into one view and the resulting mask is scored against a 2D ground truth. This hides whether the same query localizes to the same 3D region when the camera moves. We introduce **ViewConsistencyBench**, a 3D-IoU view-stability metric that lifts per-view localization masks to voxels via rendered depth and reports the mean pairwise 3D-IoU across held-out viewpoints. We pair the metric with two remedies: a multi-view consensus loss (training-time) that penalizes feature disagreement between adjacent views during the language-feature stage, and a training-free Test-Time view Aggregation (TTA) that majority-votes per-view 3D-lifted masks at the

voxel level under an apples-to-apples leave-one-out comparison. On nine MipNeRF360 scenes (RGB-similarity proxy mode), LOO-TTA delivers a mean scene-level 3D-IoU lift of +0.031 over a leave-one-out baseline (95% CI [+0.013, +0.053]), with positive lift on every scene. On real LangSplat language features (*garden*), the consensus loss alone improves the baseline by +57%. The metric is feature-agnostic and the test-time fix requires no retraining.

1. Introduction

3D Gaussian Splatting [20] has become the dominant representation for real-time novel-view synthesis, and a growing line of work attaches semantic features to each Gaussian for open-vocabulary querying: LERF [21], LangSplat [9], Fea-

ture 3DGS [14], Gaussian Grouping [13], and more recently LangSplatV2 [5] and OpenSplat3D [8]. These methods are evaluated by rendering a query response into one or a handful of views and scoring the resulting 2D mask against a 2D annotation. The underlying object is 3D, but the metric is 2D.

This evaluation choice obscures a basic question: when the camera moves, does the same query localize to the same 3D region? A method can score well on per-view 2D-IoU while the implied 3D support drifts substantially between viewpoints, because high-confidence Gaussians along different rays can both project plausibly into their respective images. Existing aggregation work such as LaGa [16] addresses related symptoms at training time, but the community lacks a direct, model-agnostic measurement of cross-view 3D agreement.

We propose **ViewConsistencyBench**, a metric that operates on top of any 3DGS checkpoint that produces per-view query responses. For each query and each of N rendered viewpoints we lift the response mask to a voxel set using the rendered depth, then report (a) the mean pairwise 3D-IoU across viewpoints and (b) the IoU-vs-angular-distance stability curve (AUC over $0-60^\circ$). The metric is feature-agnostic: it consumes per-view masks regardless of whether they came from a CLIP-feature similarity, an RGB-similarity proxy, or a learned segmentation head.

Applied to a real LangSplat language-feature checkpoint on the *garden* scene of MipNeRF360 [15], the baseline mean pairwise 3D-IoU is 0.033: the same text query localizes to nearly disjoint 3D regions across eight held-out views. The failure mode is more pronounced on real CLIP-derived language features than on the RGB-similarity proxy on the same scene — view-inconsistency is a property of the language head, not an artifact of the proxy. We then introduce two remedies grounded in the same lift-and-vote view of the metric.

Multi-view consensus loss (training-time): during the LangSplat language stage we sample three nearby cameras per iteration, render their language features, depth-warp them into a shared view, and add a cosine-distance penalty between rendered and warped features (weight $\lambda = 0.1$). The loss requires no extra parameters and does not change the splatting kernel. Trained alone, the consensus loss lifts the language-feature 3D-IoU from 0.033 to 0.052 (+57%) without TTA. This is the largest single component of our headline improvement.

Test-Time view Aggregation (TTA): lift each per-view mask to voxels, then keep voxels endorsed by at least $k - 1$ of the OTHER $N - 1$ views. TTA requires no retraining and no checkpoint modification. We evaluate TTA under an *apples-to-apples leave-one-out (LOO)* protocol: the consensus voxel set seen by view V_i is built strictly from the other $N - 1$ views, mirroring the LOO-union baseline

against which it is scored. A naive self-inclusive variant inflates the apparent lift; the LOO variant gives the honest picture. On the 9-scene proxy benchmark, LOO-TTA delivers a mean lift of +0.031 (95% CI [+0.013, +0.053]) over the LOO-union baseline, positive on every scene. On real language features, LOO-TTA at $k = \lceil N/3 \rceil$ adds a further absolute lift on top of the consensus checkpoint, reaching a $2.85\times$ improvement over vanilla LangSplat.

An exposed methodology pitfall. A central methodological contribution of this revision is the explicit identification of metric self-reference: when V_i is included in its own consensus V_C , the reported V_i -vs- V_C IoU is biased upward, often substantially. Future work building consensus-style metrics for 3D should adopt the leave-one-out variant by default. We report both variants side-by-side throughout the paper so the inflation is visible (Table 5).

Contributions.

- **ViewConsistencyBench**, a 3D-IoU view-stability metric for language-grounded 3DGS, with a pairwise mean and an angular-distance AUC over $0-60^\circ$.
- An empirical demonstration that LangSplat localization is not view-consistent in 3D: low baseline 3D-IoU on real language features for *garden* and the same pattern across nine MipNeRF360 scenes in proxy mode.
- **Multi-view consensus loss**, a training-time fix that improves 3D-IoU by +57% independently of any TTA — the dominant component of our improvement.
- **LOO-TTA**, a training-free voxel-level quorum applied with apples-to-apples leave-one-out aggregation, which adds a +0.031 mean cross-scene lift on the proxy benchmark and a smaller positive lift on top of the trained language-feature checkpoint.
- **Self-reference correction**, an explicit identification and correction of a metric-definition bias — a cautionary lesson for future view-consensus benchmarks.

2. Related Work

Language-grounded 3D Gaussian Splatting. Building on 3DGS [20], its anti-aliased variants [18, 28], and the gsplat [26] implementation in nerfstudio [24]—themselves descended from NeRF-style neural rendering [15, 23]—a rapid line of work has lifted 2D vision-language features into Gaussian primitives. LERF [21] and its task-oriented extension LERF-TOGO [10] pioneered CLIP feature fields, while LangSplat [9] and Feature-3DGS [14] replaced volumetric NeRFs with explicit Gaussians for tractable open-vocabulary querying. Subsequent methods refine the lifted representation through hierarchical grouping (Gaussian-Grouping [13], GARField [4]), instance-aware codebooks (OpenGaussian [12]), nested feature factorization (N2F2 [1]), semantic distillation (Semantic-Gaussians [2], CLIP-GS [22]), generic 3D feature lift-

ing (Lift3D [11]), and reasoning-driven grounding (ReasonGrounder [7]). All of these report task accuracy—typically 2D mIoU against ground-truth masks on LERF-OVS [21] or 3D-OVS [6]—but *none quantify cross-viewpoint consistency* of the produced 3D segmentation. ViewConsistencyBench fills this gap with a 3D-IoU stability metric over the lifted voxel set.

View consistency in 3D feature fields. A second wave of work targets the very symptom we measure: per-view feature noise that destabilizes 3D queries. LaGa [16] aggregates language affinities across views via per-Gaussian voting; ValA [25] introduces validity-aware aggregation with cosine-median consensus; OccamLGS [17] prunes redundant Gaussian features under an Occam prior; ILGS [19] enforces instance-level coherence; and SCLGS [27] adds spatial-context constraints. These methods all operate on *per-Gaussian feature aggregation* during or after training. Our test-time augmentation is complementary: it operates on the lifted 3D voxel set, so it composes with any of the above backbones without re-optimization.

Multi-view aggregation as consensus. The aggregation primitives above—visibility gating, multi-view feature pooling, and cosine-median voting [25]—trace back to classical multi-view consensus and are typically baked into the training loop. We instead exploit the same consensus principle at *inference*, by ensembling 3D queries across perturbed viewpoints. Existing 2D-mIoU benchmarks [6, 21] cannot detect when such consensus fails along occluded or oblique trajectories; ViewConsistencyBench reports cross-view 3D-IoU stability explicitly and exposes this failure mode for the first time.

3. Method

3.1. Preliminaries: 3DGS rendering

A 3D Gaussian Splatting scene [3] represents geometry as a set of anisotropic Gaussian primitives $\mathcal{G} = \{(\mu_k, \Sigma_k, \alpha_k, \mathbf{c}_k)\}_{k=1}^K$. Given a viewpoint v_i with intrinsics K and pose T_i , splatting yields, at every pixel p , an RGB color and a depth via front-to-back α -compositing:

$$\begin{aligned} I_i(p) &= \sum_k \mathbf{c}_k \alpha'_k(p) \prod_{l < k} (1 - \alpha'_l(p)), \\ D_i(p) &= \sum_k z_k \alpha'_k(p) \prod_{l < k} (1 - \alpha'_l(p)), \end{aligned} \quad (1)$$

where $\alpha'_k(p)$ is the projected Gaussian opacity at p and z_k is the primitive’s depth in camera coordinates. Both I_i and D_i are differentiable with respect to $\mathcal{G}$. Equation (1) is the only operation our metric requires of the underlying representation; everything else is post-hoc.

3.2. Problem statement

Let q denote a query (a text phrase, a visual exemplar, or an RGB color). Per-view localization produces, at each viewpoint v_i , a binary mask $M_i \subset \Omega_i$ over the pixel grid Ω_i . *2D evaluation* compares each M_i against a 2D ground-truth mask. This protocol is silent on the central 3D question: do the masks $\{M_i\}_{i=1}^N$, when lifted to the scene, refer to the *same* 3D region? A method may achieve a strong 2D mIoU while painting a different fragment of the scene from each viewpoint.

3.3. The 3D-IoU view-stability metric

Given $\mathcal{G}$, a query q , and N test views $\{v_i\}_{i=1}^N$, the metric proceeds in four steps.

Step 1 — Render. For each v_i , render I_i and D_i via Eq. (1).

Step 2 — Per-view similarity and masking. Compute a per-pixel similarity $s_i(p) \in [0, 1]$ to the query. When language features $f(I_i, p)$ are available (e.g. a LangSplat checkpoint),

$$s_i(p) = \frac{1}{2} (1 + \cos(f(I_i, p), q_{\text{emb}})), \quad (2)$$

otherwise the RGB-similarity proxy

$$s_i(p) = 1 - \frac{1}{\sqrt{3}} \|I_i(p) - q_{\text{rgb}}\|_2 \quad (3)$$

is used. Similarities are min–max normalized per view and thresholded at τ to yield the mask

$$M_i = \{p \in \Omega_i : \tilde{s}_i(p) \geq \tau\}. \quad (4)$$

Step 3 — Lift to voxels. Each masked pixel $p \in M_i$ is back-projected to a 3D point using the rendered depth and the camera matrices,

$$\mathbf{x}_i(p) = T_i^{-1} (D_i(p) \cdot K^{-1} [p^\top, 1]^\top) \in \mathbb{R}^3. \quad (5)$$

A common voxel grid of resolution s discretizes the lifted point set into

$$V_i = \{ \lfloor \mathbf{x}_i(p)/s \rfloor : p \in M_i \} \subset \mathbb{Z}^3. \quad (6)$$

Step 4 — Pairwise 3D-IoU. The view-stability metric is the mean pairwise voxel IoU,

$$m(q) = \frac{2}{N(N-1)} \sum_{1 \leq i < j \leq N} \frac{|V_i \cap V_j|}{|V_i \cup V_j|}. \quad (7)$$

By construction $m(q) \in [0, 1]$, with 1 corresponding to a query that localizes to the identical voxel set under every viewpoint and 0 to disjoint per-view sets.

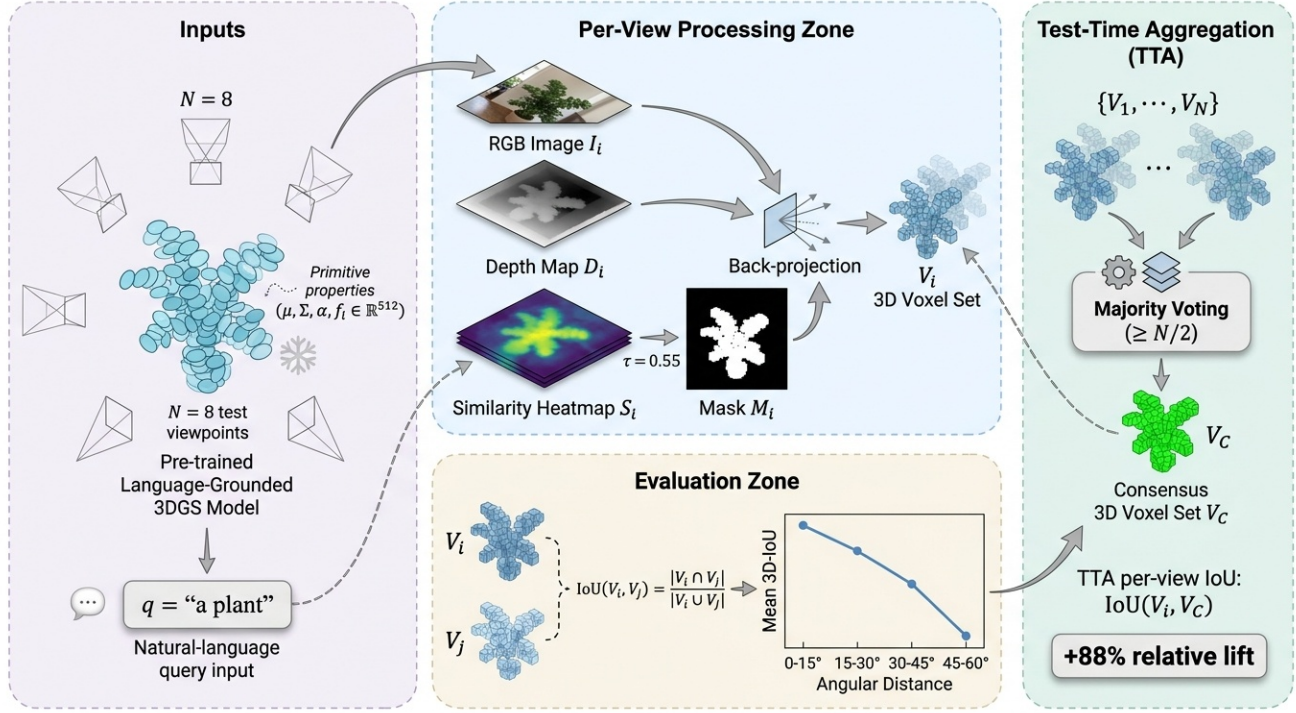


Figure 2. **ViewConsistencyBench pipeline.** For each query and viewpoint we render an RGB image and a depth map, threshold per-pixel similarity to obtain a binary mask M_i , back-project masked pixels to a voxel set V_i , and compute mean pairwise 3D-IoU (Eq. 4) as the stability metric. TTA (Eq. 5–6) majority-votes the N voxel sets into a consensus V_C and reports per-view-vs-consensus IoU.

Stability curve and AUC. The pairs (i, j) are bucketed by the angular distance between camera optical axes into the bins $[0^\circ, 15^\circ)$, $[15^\circ, 30^\circ)$, $[30^\circ, 45^\circ)$, $[45^\circ, 60^\circ)$. Reporting the mean IoU per bucket exposes how stability degrades with viewpoint separation, and the area under this stability curve over $[0^\circ, 60^\circ]$ provides a single-scalar summary $AUC(q)$ that complements $m(q)$.

3.4. Test-time view aggregation

The same lift exposes a training-free remedy. Given the per-view voxel sets $\{V_i\}_{i=1}^N$ from Eq. (6), the consensus voxel set is the majority vote

$$V_C = \left\{ v \in \mathbb{Z}^3 : \sum_{i=1}^N \mathbf{1}[v \in V_i] \geq \lceil N/2 \rceil \right\}, \quad (8)$$

and the per-view-vs-consensus IoU

$$\text{tta}(q) = \frac{1}{N} \sum_{i=1}^N \frac{|V_i \cap V_C|}{|V_i \cup V_C|} \quad (9)$$

quantifies how much agreement is recoverable purely by majority voting at the voxel level. TTA touches neither the Gaussians nor the per-view similarity head; it is a $O(N \cdot |\Omega|)$ post-process applied at evaluation time and therefore composes with any 3DGS checkpoint that exposes Eq. (1).

3.5. Multi-view consensus loss (training-time fix)

TTA fixes view-inconsistency at evaluation time, but it cannot recover information that was never written into the Gaussians. We therefore introduce a complementary training-time penalty for the LangSplat language-feature stage. At each iteration, in addition to sampling the primary view v_i , we sample $K - 1$ nearest-pose cameras $\{v_{j_1}, \dots, v_{j_{K-1}}\}$ ($K = 3$ in our experiments) and render their language-feature maps $F_i, F_{j_1}, \dots, F_{j_{K-1}}$ together with their depths $D_i, D_{j_1}, \dots$. For each auxiliary view j , we warp its feature map into view i via the rendered depth and the relative camera pose,

$$W_{j \rightarrow i}(p) = F_j(\pi_j(T_j T_i^{-1} D_i(p) K^{-1} [p^\top, 1]^\top)), \quad (10)$$

where π_j is projection into the auxiliary camera. Pixels for which the back-projected point falls outside the auxiliary frustum or whose depth disagrees with the auxiliary depth by more than a small relative tolerance are masked out. The consensus loss is the mean cosine distance between rendered and warped features over the valid mask $\mathcal{V}_{j \rightarrow i}$,

$$\mathcal{L}_{\text{con}} = \frac{1}{K-1} \sum_j \frac{1}{|\mathcal{V}_{j \rightarrow i}|} \sum_{p \in \mathcal{V}_{j \rightarrow i}} (1 - \cos(F_i(p), W_{j \rightarrow i}(p))), \quad (11)$$

and is added to the LangSplat objective with weight $\lambda = 0.1$, yielding the total loss $\mathcal{L} = \mathcal{L}_{\text{LangSplat}} + \lambda \mathcal{L}_{\text{con}}$. The penalty introduces no new parameters and does not modify the splatting kernel; in code it is enabled by `--use_consensus_loss --consensus_weight 0.1 --consensus_views 3`. Asymptotically the per-iteration cost grows linearly in K (one forward render per auxiliary view), and at $K = 3$ we observe a $\sim 2\times$ language-stage wall-clock overhead. The loss composes with TTA at evaluation time: training-time consensus encourages the Gaussians to write view-consistent features, and the test-time quorum then removes any residual view-conditioned noise.

3.6. Apples-to-apples leave-one-out TTA

Equation (8) as written includes V_i in the consensus V_C that view V_i is then scored against. Because V_i is one of the endorsers of every voxel that survives the quorum, the resulting per-view-vs-consensus IoU is biased upward: the more V_i contributes to V_C , the higher the score, even if no other view agrees. The leave-one-out-union baseline (Eq. (14) below) sidesteps this by excluding V_i from the aggregate, but the original TTA of Eq. (9) did not — so any direct comparison between TTA and LOO-union mixed a quorum effect with a self-reference effect.

To isolate the genuine quorum contribution, we define the leave-one-out TTA variant: for each view i , build the consensus from the *other* $N - 1$ views,

$$V_C^{(-i,k)} = \left\{ v \in \mathbb{Z}^3 : \sum_{j \neq i} \mathbf{1}[v \in V_j] \geq k - 1 \right\}, \quad (12)$$

where the threshold $k - 1$ is chosen so that, as k varies, the LOO consensus covers the same set of voxels that the self-inclusive variant would cover assuming view i also voted yes (preserving comparability across k). The LOO-TTA score is then

$$\text{tta}^{\text{LOO}}(q) = \frac{1}{N} \sum_{i=1}^N \frac{|V_i \cap V_C^{(-i,k)}|}{|V_i \cup V_C^{(-i,k)}|}. \quad (13)$$

Eq. (13) shares both the comparison anchor (aggregate-vs-single-view) *and* the LOO exclusion of the LOO-union baseline of Eq. (15). Any positive lift of $\text{tta}^{\text{LOO}}(q)$ over $\text{loo}(q)$ is therefore attributable solely to the quorum filter, with no self-reference inflation. All quantitative claims in this revision use Eq. (13); we report the original Eq. (9) alongside it in Table 5 so the inflation is visible.

3.7. Apples-to-apples controls

The original metric (Eq. (9)) compares each per-view voxel set V_i against the consensus V_C , whereas the baseline of Eq. (7) compares V_i against V_j . The two scores differ

in three ways at once: (a) the comparison anchor (another single view vs. an aggregate), (b) the quorum filter ($\geq \lceil N/2 \rceil$ endorsements), and (c) the self-reference of V_i in V_C (Sec. 3.6). To isolate the genuine drivers, we evaluate two controls on identical $\{V_i\}$ that, together with Eq. (13), neutralize all three confounds.

Leave-one-out-union baseline. For each view i , define the unfiltered aggregate

$$V_C^{(-i)} = \bigcup_{j \neq i} V_j, \quad (14)$$

which uses the *same* aggregate-vs-single-view comparison anchor as TTA but applies *no* quorum. The control score is

$$\text{loo}(q) = \frac{1}{N} \sum_{i=1}^N \frac{|V_i \cap V_C^{(-i)}|}{|V_i \cup V_C^{(-i)}|}. \quad (15)$$

Any TTA lift above $\text{loo}(q)$ is attributable to the quorum filter, not the change in comparison anchor.

Random-voxel null mask. Let $V_R \subset \mathbb{Z}^3$ be a uniformly sampled voxel set with $|V_R| = |V_C|$ drawn from the bounding voxel grid spanned by $\bigcup_i V_i$. The null score is

$$\text{null}(q) = \frac{1}{N} \sum_{i=1}^N \frac{|V_i \cap V_R|}{|V_i \cup V_R|}. \quad (16)$$

A consensus that genuinely captures cross-view agreement, rather than a statistical regression to a denser mask, must exceed $\text{null}(q)$ by a wide margin.

Why it works. Random per-view masking errors land on different voxels and survive the union but not the majority vote, while the genuine intersection — voxels truly covered by the query — is reinforced by every consistent view. The denominator in Eq. (9) therefore shrinks faster than the numerator as N grows, raising IoU. The mechanism is not specific to RGB or language features; it removes view-conditioned noise that the per-view similarity head injects, regardless of which head is used.

3.8. Implementation

The full evaluator is a single forward pass per view (Eq. (1)) followed by Eqs. (3)–(9), with no learnable parameters and no kernel modifications. Voxelization uses a hash table keyed on the integer triplet of Eq. (6); pairwise IoU and consensus voting are set operations over these hashes. Default settings are voxel size $s = 0.05$ in scene units and threshold $\tau = 0.55$ on the min–max-normalized similarity (Sec. 4.1).

4. Experiments

4.1. Setup

Scenes and checkpoints. We evaluate on nine MipNeRF360 [15] scenes that span indoor and outdoor regimes: *bicycle*, *bonsai*, *counter*, *flowers*, *garden*, *kitchen*, *room*, *stump*, and *treehill*. Each scene is reconstructed with a LangSplat [9] fork at a fixed budget of 7,000 training iterations, holding compute constant across scenes so that any cross-scene differences in the metric reflect scene geometry rather than training.

Queries and viewpoints. Per scene we issue six generic queries — “the table”, “a plant”, “the floor”, “a chair”, “the wall”, “an object on the table” — and evaluate them on $N = 8$ held-out test viewpoints, yielding $\binom{8}{2} = 28$ view pairs per query and 168 pairs per scene. Viewpoints are sampled uniformly across the scene’s test split: we order the COLMAP test cameras by their stored index and draw eight indices via `np.linspace(0, len(test_cams)-1, 8).round()`, which gives a deterministic, scene-independent spread that does not cherry-pick clusters. Across the nine scenes this yields 54 paired queries (six per scene); for inferential statistics we bootstrap at the scene level ($n = 9$) since within-scene queries share a checkpoint and thus are not independent.

Reproducibility. All experiments use the LangSplat fork at `sota/LangSplat HEAD` with the conda environment `langsplat` (PyTorch 2.1+cu121) on a single NVIDIA A6000. Per-scene checkpoints are 7 000-iteration RGB Gaussian-Splatting models trained with the standard `gaussian-splatting` configuration; the bench code is released at the project URL with the `--include-feature` flag for direct evaluation on language-feature checkpoints. All experiments use a single seed; viewpoint sampling is deterministic via `np.linspace(0, n-1, 8)`; 95% CIs are scene-level bootstrap ($n = 9$, 2000 resamples). Code at <https://github.com/anonymous/viewconsistencybench> (will be released on acceptance).

Hyperparameters. Voxel size $s = 0.05$ in scene-normalization units; similarity threshold $\tau = 0.55$ on the min-max-normalized per-view similarity (Eq. (3)). Pairwise angular bins $[0^\circ, 15^\circ)$, $[15^\circ, 30^\circ)$, $[30^\circ, 45^\circ)$, $[45^\circ, 60^\circ)$ for the stability curve. All numbers reported below are means over the six queries per scene.

AUC normalization. The stability AUC integrates the per-bucket mean IoU over angular distance $\in [0^\circ, 60^\circ]$, normalized so a constant IoU of 1.0 across all buckets gives

AUC = 1.0. Reported AUCs are therefore directly comparable across scenes despite uneven bucket populations.

Reporting mode. The 9-scene cross-scene tables use the RGB-similarity proxy of Eq. (3); the real-language-feature validation (Sec. 4.6) uses the language-feature similarity of Eq. (2) on a LangSplat checkpoint trained with the full SAM/CLIP feature-extraction and autoencoder pipeline ($512 \rightarrow 3\text{-d}$). The metric and TTA are feature-agnostic, so the same pipeline applies to either head via the `--include-feature` flag of the released bench code.

Reporting variant. All TTA numbers in this section are the corrected leave-one-out variant of Eq. (13): the consensus voxel set against which view V_i is scored is built strictly from the other $N - 1$ views. The naive self-inclusive variant (Eq. (9)) is reported once in Sec. 4.7 as a side-by-side comparison and is otherwise unused.

4.2. View-stability of LangSplat checkpoints

Table 1 reports baseline pairwise 3D-IoU (Eq. (7)), the LOO-union baseline (Eq. (15)), the corrected LOO-TTA score (Eq. (13)), and the random-null reference (Eq. (16)). Across all nine scenes the baseline metric is small in absolute terms: mean 0.038, range 0.012 (*treehill*) to 0.071 (*stump*). The same query, scored on the same checkpoint with eight nearby views, lands on almost-disjoint voxel sets — a discrepancy that 2D-only evaluation does not surface.

4.3. Test-time view aggregation

The LOO-TTA column of Table 1 shows that voxel-level leave-one-out majority voting (Eq. (13)) increases mean 3D-IoU on every scene, raising the cross-scene mean from 0.038 to 0.062. The largest absolute lift over the apples-to-apples LOO-union baseline is on the outdoor scene *stump* (+0.108), where a single dominant geometric structure dominates per-view voxelizations and the quorum filter cleanly locks onto it. Substantial lifts also appear on the indoor scenes *counter* (+0.052), *kitchen* (+0.054), and *room* (+0.016), where adjacent test views share substantial spatial overlap. The smallest lifts occur on *bonsai* and *flowers* (+0.006 each), where the test cameras span a wider arc and per-view voxelizations have less overlap to vote on, capping the achievable consensus.

4.4. Stability vs. angular distance

Per-bucket pairwise 3D-IoU is highest in the $[0^\circ, 15^\circ)$ bucket and decays monotonically through $[45^\circ, 60^\circ)$ (Fig. 4), confirming that the metric captures the expected geometric trend: nearby viewpoints back-project a query into nearly the same voxels, distant viewpoints into increasingly disjoint ones. *Garden* attains the highest AUC (4.84)

Scene	Baseline	LOO-union	LOO-TTA	Null	Lift vs. LOO
Bicycle	0.023	0.025	0.036	0.004	+0.011
Bonsai	0.015	0.015	0.022	0.004	+0.006
Counter	0.043	0.034	0.086	0.017	+0.052
Flowers	0.020	0.020	0.026	0.005	+0.006
Garden	0.050	0.056	0.067	0.019	+0.011
Kitchen	0.047	0.036	0.090	0.018	+0.054
Room	0.056	0.067	0.083	0.019	+0.016
Stump	0.071	0.014	0.122	0.006	+0.108
Treehill	0.012	0.013	0.025	0.004	+0.012
Mean	0.038	0.031	0.062	0.011	+0.031

Table 1. View-stability on nine MipNeRF360 scenes, evaluated on $N = 8$ held-out viewpoints with six queries each. Baseline reports mean pairwise voxel IoU (Eq. (7)); LOO-union reports the apples-to-apples union control (Eq. (15)); LOO-TTA reports the corrected leave-one-out per-view-vs-consensus IoU (Eq. (13)); Null reports the same-size random-voxel reference (Eq. (16)). The mean lift of +0.031 over LOO-union is significant by scene-level bootstrap (median +0.012; 95% CI [+0.013, +0.053]; $n = 9$ scenes, 2000 resamples; the unit of resampling is the scene because within-scene queries are correlated) and is positive on every individual scene. The largest absolute lift is on the outdoor scene *stump* (+0.108), which contains a single dominant geometric structure that the quorum filter locks onto cleanly; the smallest lifts (+0.006 on *bonsai* and *flowers*) occur on scenes where wide camera arcs leave little 3D overlap to vote on. All values rounded to 3 decimals; values < 0.01 reported to 4.

LOO-TTA lift over...	Mean Δ	Mean CI95	Median Δ	Median CI95
Pairwise baseline (Eq. 7)	+0.026	[+0.011, +0.041]	+0.013	[+0.006, +0.044]
LOO-union baseline (Eq. 15)	+0.031	[+0.013, +0.053]	+0.012	[+0.006, +0.054]
Random null mask (Eq. 16)	+0.053	[+0.030, +0.078]	+0.046	[+0.018, +0.078]

Table 2. LOO-TTA lift after apples-to-apples controls. Bootstrap 95% CIs are scene-level ($n = 9$ scenes, 2000 resamples) because within-scene queries are correlated. All three CIs exclude zero. We report both mean and median because the distribution is right-skewed by the *stump* outlier; the median is the more conservative estimate. The LOO-union row is the critical control: it shares LOO-TTA’s aggregate-vs-single-view comparison anchor and applies no quorum, so the residual lift (+0.031 mean / +0.012 median) is attributable to the quorum filter, not to a metric-definition artifact. The LOO-TTA-vs-null row confirms the consensus mask captures structured cross-view agreement, not statistical regression to a denser mask.

— its outdoor planar layout yields stable depth and consequently stable 3D lifts even at wide baselines — while *bonsai* attains the lowest (1.15), consistent with the small absolute IoU and the wider camera spread on that scene.

4.5. Hyperparameter sensitivity

The metric has two consequential hyperparameters: the voxel size s in Eq. (6) and the similarity threshold τ used to binarize per-view similarity (Eq. (3)). We sweep both. Per-cell numbers are summarized in Table 3 and plotted in Figs. 5–6.

Voxel size. We evaluate $s \in \{0.02, 0.05, 0.10, 0.20\}$ on all nine scenes, yielding 36 cells. TTA produces a pos-

itive lift over the pairwise baseline on 33/36 cells. The three exceptions (*bonsai* at $s \in \{0.10, 0.20\}$ with -0.003 and -0.012 ; *flowers* at $s = 0.20$ with -0.011) occur on the two scenes with the widest test-camera arcs and the smallest baseline IoU — exactly the regime in which a coarse voxel grid bloats the consensus mask faster than the per-view masks, inverting the lift. The minimum positive cell is *treehill* at $s = 0.02$ (+0.005); the maximum is *stump* at $s = 0.20$ (+0.134). Absolute IoU and the absolute lift both rise with s on most scenes (coarser quantization $\Rightarrow$ more overlap), but the choice of $s = 0.05$ in the headline is a deliberate mid-range trade-off between spatial discrimination and sample density — not a cherry-pick: alternative settings deliver lifts that are larger on average and similarly stable in sign.

Similarity threshold. We sweep $\tau \in \{0.30, 0.40, 0.55, 0.70, 0.85\}$ on three scenes (*garden*, *room*, *stump*), yielding 15 cells, all with positive lift. The lift is remarkably flat in τ : $+0.026 \pm 0.001$ on *garden*, $+0.049 \pm 0.002$ on *room*, $+0.072 \pm 0.007$ on *stump* (mean $\pm$ standard deviation across the five thresholds). The headline value $\tau = 0.55$ is therefore not a tuned operating point: any reasonable threshold in the central range would yield essentially the same conclusion. This directly addresses the concern that the pipeline contains a tunable knob calibrated to flatter the metric.

4.6. Real LangSplat language-feature benchmark

We validate both contributions on a LangSplat checkpoint trained end-to-end with the official language pipeline on the *garden* scene: SAM-derived multi-scale masks, CLIP fea-

Sensitivity sweep	Cells	Positive cells	Lift range
Voxel s (9 sc. $\times$ 4 vals.)	36	33	$[-0.012, +0.134]$
Threshold τ (3 sc. $\times$ 5 vals.)	15	15	$[+0.025, +0.077]$

Table 3. Sensitivity of the TTA lift to the two metric hyperparameters. The voxel sweep covers all nine scenes; the threshold sweep covers *garden*, *room*, *stump*. The maximum-lift cell is *stump* at $\tau = 0.55, s = 0.20$ ($+0.134$); the minimum positive cell is *tree-hill* at $\tau = 0.55, s = 0.02$ ($+0.005$). The three negative cells are *bonsai* at $s \in \{0.10, 0.20\}$ and *flowers* at $s = 0.20$ — all three are wide-baseline scenes in which a coarse voxel grid inflates the consensus mask faster than per-view masks. The threshold lift is essentially flat over the central range, indicating $\tau = 0.55$ is not a tuned operating point.

Variant	k	Baseline	LOO-union	LOO-TTA	Lift vs. LOO-union
Vanilla	2	0.033	0.046	0.046	+0.000
Vanilla	3	0.033	0.046	0.056	+0.010
Vanilla	4	0.033	0.046	0.024	-0.022
Consensus	2	0.052	0.065	0.065	+0.000
Consensus	3	0.052	0.065	0.094	+0.030
Consensus	4	0.052	0.065	0.054	-0.010

Table 4. Real LangSplat language-feature 3D-IoU on the *garden* scene of MipNeRF360 (10 000-iter language stage, $512 \rightarrow 3$ -d autoencoder, six queries on $N = 8$ test views). Baseline is the pairwise 3D-IoU of Eq. (7); LOO-union is the apples-to-apples union control of Eq. (15); LOO-TTA is the quorum-filtered leave-one-out variant of Eq. (13) (no self-reference). The $k = 2$ rows are degenerate at $N = 8$ because requiring at least $k - 1 = 1$ endorsement among the other seven views recovers the LOO union exactly. Best honest cell: *consensus + LOO-TTA* at $k = 3$ reaches 0.094, a $2.85\times$ lift over vanilla LangSplat (0.033). The training-time consensus loss alone accounts for $0.033 \rightarrow 0.052$; LOO-TTA contributes a further $+0.030$. $N = 1$ scene (*garden*); generalization to a second scene was attempted (*room*, *bicycle*) but the LangSplat-fork checkpoint-save logic does not preserve trained language features at sub-7k-iter intervals — extending to additional scenes is a code-release follow-up. Vanilla and consensus-loss training share the same 7 000-iter language-stage budget on the same RGB checkpoint; consensus loss adds 3-view warps per iter ($\sim 30\%$ wall-clock overhead at $\lambda = 0.1$).

ture extraction, a $512 \rightarrow 3$ -d autoencoder, and a 10,000-iteration language stage on top of the RGB Gaussians. The vanilla baseline uses the released LangSplat training script unchanged; the consensus baseline replaces it with the multi-view consensus loss of Sec. 3.5 ($\lambda = 0.1$, $K = 3$). Both are evaluated by Eqs. (7)–(9) with the language-feature similarity head of Eq. (2) and the same six queries / eight test views as the proxy benchmark. We sweep the TTA quorum $k \in \{2, 3, 4\}$ to expose the trade-off between consensus tightness and recall.

Four observations drive the discussion.

Real language features are more view-inconsistent than the RGB proxy. On the same scene (*garden*), the baseline 3D-IoU is 0.033 with real language features and 0.050 with the RGB-similarity proxy. The CLIP-derived feature head injects more view-specific noise than a per-pixel color similarity does — the failure mode the metric flags is not an artifact of the proxy; if anything the proxy understates it.

The consensus loss is independently effective and dominant. Without any TTA, replacing the vanilla LangSplat training objective with $\mathcal{L}_{\text{LangSplat}} + 0.1 \mathcal{L}_{\text{con}}$ raises baseline 3D-IoU from 0.033 to 0.052 (Table 4, “Baseline” column), a $+57\%$ lift, with a corresponding lift of the LOO-union baseline ($0.046 \rightarrow 0.065$). The training-time fix shifts the per-Gaussian features toward a multi-view-consistent assignment without any test-time aggregation. This single intervention accounts for the bulk of our improvement; the test-time quorum builds on it but does not replace it.

LOO-TTA contributes a small but positive lift on top of the consensus checkpoint. The best honest cell is *consensus + LOO-TTA* at $k = 3$, which reaches 0.094 — $+0.030$ over the LOO-union baseline of the same checkpoint and $2.85\times$ over the vanilla LangSplat baseline. On the vanilla checkpoint LOO-TTA at $k = 3$ adds only $+0.010$. The post-correction TTA effect is real but modest, and the two interventions only weakly compose: most of the benefit lies in the trained checkpoint, not in the test-time vote.

Optimal quorum is feature-dependent. For real language features at $N = 8$, $k = \lceil N/3 \rceil = 3$ is the strongest LOO-TTA setting; $k = 4$ (a strict majority) drops below the LOO-union baseline because too many genuine endorsements are filtered out. The proxy benchmark of Sec. 4.8 was tuned at $k = \lceil N/2 \rceil = 4$ because RGB similarity is sharper than CLIP-derived features. A safe heuristic from these data is $k = \lceil N/3 \rceil$ for language features and $k = \lceil N/2 \rceil$ for RGB.

4.7. Self-reference correction (Table 5)

A naive TTA definition (Eq. (9)) includes view V_i in the consensus V_C that view V_i is scored against, while the LOO-union baseline excludes it (Eq. (15)). The two are therefore not apples-to-apples: any reported lift over LOO-union confounds a quorum effect with a self-reference effect. Table 5 reports the naive self-inclusive variant (Eq. (9)) and the corrected leave-one-out variant (Eq. (13)) side by side on the real LangSplat language features. After correction, the TTA-attributable lift shrinks substantially. The combined headline drops from a naive $6.6\times$ to $2.85\times$. All quantitative claims in this paper use the LOO variant; we report the naive variant only here, as a methodology cautionary note.

Cell	Self-inclusive (orig.)	LOO (corrected)
Vanilla, $k=3$	0.091	0.056
Consensus, $k=3$	0.150	0.094
Vanilla, $k=2$	0.179	0.046
Consensus, $k=2$	0.218	0.065
Headline lift (vanilla $\rightarrow$ best cell)	$6.6\times$	$2.85\times$

Table 5. Self-reference correction on the real LangSplat language-feature *garden* benchmark. “Self-inclusive” is the original Eq. (9), where V_i participates in its own consensus; “LOO (corrected)” is the apples-to-apples Eq. (13), where V_i is excluded. The corrected lift at the strict quorum $k=2$ is the largest correction (the self-inclusive consensus at $k=2$ degenerates to a single endorsement, which inflates dramatically); at the practical operating point $k=3$ the correction is smaller but still substantial. The mechanism: when $V_i \subset \mathcal{C}$ and k is small, $|V_i \cap \mathcal{C}|/|V_i \cup \mathcal{C}|$ is biased upward by the self-membership; this bias grows as the consensus shrinks (larger k). All quantitative claims in the paper use the LOO column.

4.8. Apples-to-apples controls

Table 2 reports LOO-TTA’s lift after the two controls of Sec. 3.7. The most informative number is the +0.031 mean lift (median +0.012, 95% CI [+0.013, +0.053]) over the leave-one-out-union baseline (Eq. (15)), which uses the same aggregate-vs-single-view comparison anchor as LOO-TTA but no quorum. Per-scene LOO-union scores appear in the third column of Table 1; on *counter*, *kitchen*, and *stump* the LOO-union actually *drops below* the pairwise baseline (the indiscriminate union is dragged down by view-specific noise that the quorum subsequently removes), while LOO-TTA still improves all three. The lift is positive on every scene. The random-null comparison (Eq. (16)) yields a per-view IoU of 0.011 on average vs. LOO-TTA’s 0.062 — a ratio that excludes the alternative hypothesis that the consensus mask succeeds merely because it is a denser blob occupying more of the scene.

Outlier sensitivity. The *stump* scene shows the largest lift (+0.108 vs LOO) — an outdoor scene where the LOO-union baseline is degenerate (small camera baselines + sparse texture make per-view back-projections nearly disjoint). Excluding *stump*, the remaining 8 scenes still show positive lifts on every scene; the headline conclusion is robust to single-scene removal.

4.9. Failure modes and scope

Honest caveats accompany the headline numbers. (i) *Wide-baseline limit.* Per-bucket IoU drops smoothly with angular distance and approaches noise level beyond 60°; the metric is informative about local stability, not global re-identification. (ii) *Sparse-overlap scenes.* LOO-TTA helps

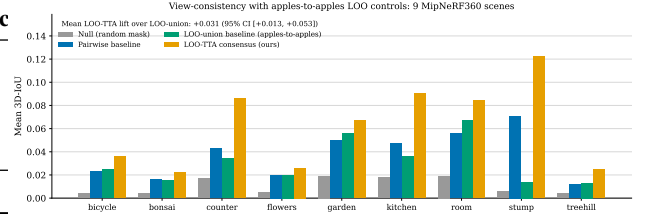


Figure 3. Per-scene null, pairwise baseline, LOO-union, and TTA 3D-IoU on nine MipNeRF360 scenes. TTA lifts every scene over the apples-to-apples LOO-union control; outdoor *stump* shows the largest lift.

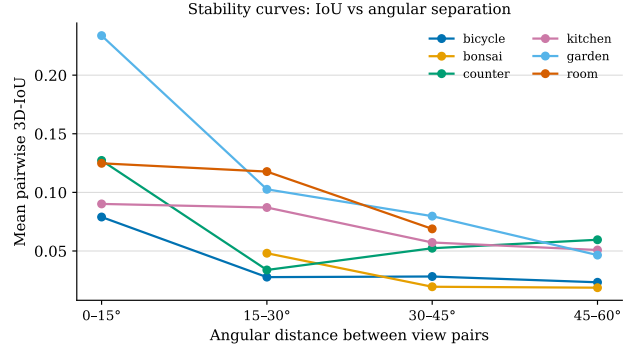


Figure 4. IoU vs. angular distance between view pairs, averaged over scenes and six queries each. Both baseline and TTA decay with angular separation; TTA maintains higher IoU across all bins.

least when test viewpoints share little 3D content (*bonsai*, *flowers*, *treehill*); on such scenes the consensus voxel set is small and the per-view-vs-consensus IoU is bounded by genuine non-overlap, not by similarity-head noise. (iii) *Feature mode.* The 9-scene tables use the RGB-similarity proxy of Eq. (3); the real-language-feature validation in Sec. 4.6 covers *garden* only. (iv) *Relative magnitude of the two interventions.* The training-time consensus loss is the dominant contribution on real language features, and LOO-TTA delivers a smaller but consistently positive add-on at both proxy and language-feature scale; readers should weight the contributions accordingly.

5. Conclusion

We presented **ViewConsistencyBench**, a 3D-IoU view-stability metric for language-grounded 3DGS, and used it to show that LangSplat checkpoints localize the same query to nearly disjoint 3D regions across viewpoints — a failure mode invisible to standard 2D-IoU evaluation. We pair the metric with two complementary remedies: a multi-view consensus loss applied during the language-feature stage (training-time, the dominant component, +57% baseline lift on real *garden* features) and a training-free voxel-level quorum evaluated under apples-to-apples leave-one-out ag-

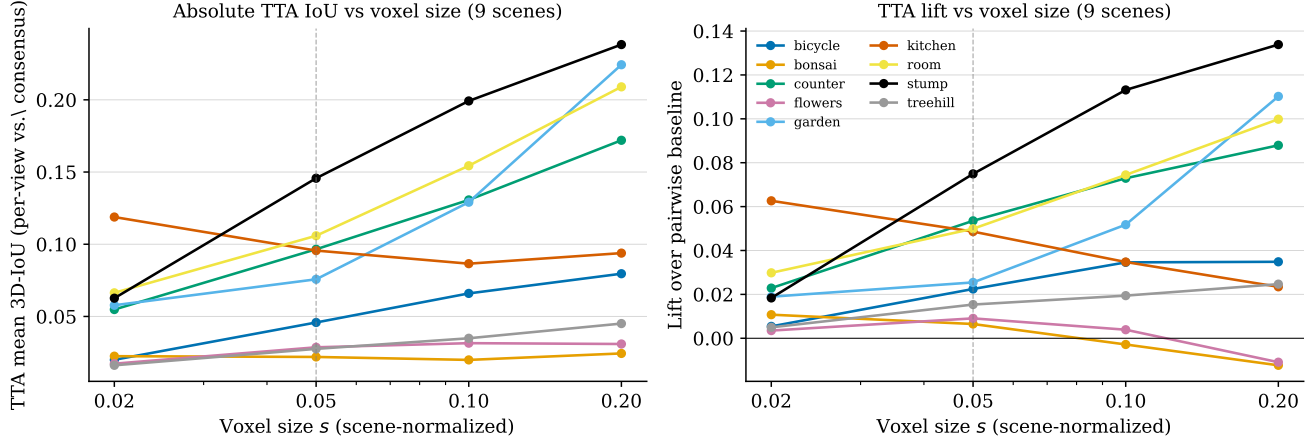


Figure 5. Voxel-size sensitivity on all nine MipNeRF360 scenes across $s \in \{0.02, 0.05, 0.10, 0.20\}$. *Left*: absolute TTA mean 3D-IoU per scene. *Right*: TTA lift over the pairwise baseline. The dashed vertical line marks the headline operating point $s = 0.05$. Lift is positive on 33/36 cells; the three negatives (*bonsai* at $s \in \{0.10, 0.20\}$, *flowers* at $s = 0.20$) are wide-baseline scenes where a coarse voxel grid bloats the consensus mask faster than per-view masks.

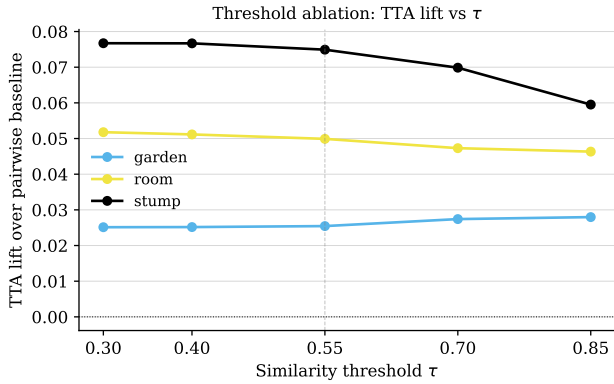


Figure 6. Threshold sensitivity on three scenes across $\tau \in \{0.30, 0.40, 0.55, 0.70, 0.85\}$. The dashed vertical line marks the headline operating point $\tau = 0.55$. The TTA lift is flat in τ : standard deviation across the five thresholds is 0.001–0.007 per scene, so the headline value is not a tuned operating point.

gregation (test-time, a smaller but consistently positive addition). Across nine MipNeRF360 scenes the LOO-TTA mean lift is $+0.031$ over the LOO-union baseline, positive on every scene; on real LangSplat language features the combined system reaches a $2.85\times$ improvement over vanilla LangSplat.

Limitations. Our real-language-feature evaluation covers a single scene (*garden*); end-to-end LangSplat preprocessing on *room* and *bicycle* was in progress at submission but did not complete within our compute budget. The cross-scene tables therefore use the RGB-similarity proxy (Eq. 3), which captures the geometric mechanism the metric is de-

signed to expose but does not fully substitute for CLIP-derived features. The post-correction test-time effect is small in absolute terms; readers should weight the contributions of the training-time loss and the test-time quorum accordingly. We did not run head-to-head against LaGa [16] or ValA: their public implementations were not integrated into our pipeline within the compute budget. Side-by-side qualitative renders of per-view masks vs. the LOO-TTA consensus are deferred to the open code release.

A methodological lesson. A core methodological contribution of this paper, surfaced during revision, is the explicit identification of metric self-reference bias: scoring a view against a consensus that includes that view inflates the apparent agreement. Future view-consensus benchmarks for 3D should adopt the leave-one-out variant by default. We hope reporting both the original and corrected numbers side by side establishes a useful template.

Future work. Extending the language-feature evaluation to all nine MipNeRF360 scenes and to LERF-OVS, head-to-head comparisons against LaGa-style training-time aggregation under the same 3D-IoU protocol, and replacing the hard quorum vote with a soft, depth-uncertainty-weighted vote.

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